RAJALAKSHMI ENGINEERING COLLEGE

Mahalakshmi B 250701401-2025-CSE

M2

Started on	Friday, 25 September 2026, 2:23 PM
State	Finished
Completed on	Friday, 25 September 2026, 2:25 PM
Time taken	1 min 28 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int main() {
    int n;
    // Read the number of elements
    if (scanf("%d", &n) != 1) {
        return 0;
    }

    // Allocate memory for the array
    int *arr = (int *)malloc(n * sizeof(int));
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int duplicate = -1;

    // Brute force search: O(n^2) time complexity, O(1) extra space
    for (int i = 0; i < n; i++) {
        for (int j = i + 1; j < n; j++) {
            if (arr[i] == arr[j]) {
                duplicate = arr[i];
                break;
            }
        }
        if (duplicate != -1) {
            break;
        }
    }

    // Print the repeated element
    printf("%d\n", duplicate);

    // Free the allocated memory
    free(arr);
    return 0;
}

```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔

	Input	Expected	Got	
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course